

DYNAMIC GAMMA-POISSON HAZARD HDP-HMM FOR ACCELERATED OBSERVATION-DEPENDENT TIME SERIES SEGMENTATION

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ABSTRACT

Bayesian nonparametric hidden Markov models, such as the Hierarchical Dirichlet Process Hidden Semi-Markov Model (HDP-HSMM) and recurrent explicit-duration models (RED-HDP-HMM), provide flexible frameworks for self-supervised time series segmentation. However, explicit duration modeling introduces an $O(TL(L + D_{\max}))$ computational bottleneck caused by the maximum duration truncation cap $D_{\max}$, alongside an $O(TLD_{\max})$ memory footprint and rigid duration sampling upon state entry that cannot update exit probabilities mid-segment. In this work, we introduce the **Dynamic Autoregressive Gamma-Poisson Hazard HDP-HMM (DAGP-HDP-HMM)**, which parameterizes state persistence as a continuous-time stochastic survival process governed by a Poisson renewal engine with a dynamic Gamma hazard rate driven by instantaneous observation feature embeddings $c_t = \phi(y_t)$. By integrating the Poisson zero-count likelihood against the Moment-Generating Function of the Gamma distribution, stay probabilities $P_{\text{stay}}(t | j)$ are evaluated in $O(1)$ closed form. This formulation bypasses $D_{\max}$ duration capping completely, reducing forward-backward state sampling complexity to $O(TL^2)$ time and $O(TL)$ memory while enabling per-frame mid-segment hazard updates. Furthermore, a soft non-loopy renewal jump kernel regularized with pseudo-count $\kappa_{\text{diag}} = 0.5$ preserves exact Dirichlet posterior conjugacy without auxiliary geometric variables. Across seven benchmark datasets, DAGP-HDP-HMM achieves state-of-the-art segmentation accuracy on Bee Dance (0.917) and highly competitive accuracy on Synthetic NASCAR (0.968, recovering the ground-truth $K = 4$ states across all seeds), yields a $17.4\times$ forward-backward speedup and $45\times$ memory reduction over RED-HDP-HMM on long sequences ($T = 10000$, $L = 100$, $D_{\max} = 500$), and demonstrates robust performance under extensive hyperparameter sensitivity and diagnostic floor analyses.

1 INTRODUCTION

Self-supervised time-series segmentation aims to partition continuous multivariate observations into discrete, semantically meaningful dynamic regimes without requiring manual annotations. Bayesian nonparametric switching state-space models—most notably the Hierarchical Dirichlet Process Hidden Markov Model (HDP-HMM) Fox et al. (2009) and the Hierarchical Dirichlet Process Hidden Semi-Markov Model (HDP-HSMM) Johnson & Willsky (2012)—provide flexible, principled tools for learning both the hidden regime sequence and an unbounded number of latent states directly from sequential data Aminikhanghahi & Cook (2016). By placing Dirichlet process priors over transition probabilities and state emissions, these frameworks bypass the need for structural model selection.

Despite their flexibility, standard HDP-HMMs inherently assume a geometric distribution over state sojourn times, which enforces a memoryless persistence prior that severely over-segments complex temporal dynamics Fox et al. (2009). To enforce realistic persistence, explicit-duration hidden semi-Markov models explicitly parameterize state duration laws using discrete distributions, such as Poisson or Negative Binomial distributions Chiappa (2014). However, explicit duration modeling introduces severe computational bottlenecks and rigid duration sampling constraints Dewar et al. (2012); Huggins & Wood (2014):

1. **Computational and Memory Bottlenecks:** Forward-backward message passing over explicit duration variables requires maintaining duration state buffers capped at a maximum length $D_{\max}$, incurring an $O(TL(L + D_{\max}))$ computational complexity per Gibbs sweep (where T is the sequence length and L is the state truncation level) and an $O(TLD_{\max})$ memory footprint Dewar et al. (2012). Setting $D_{\max}$ conservatively high to prevent truncation errors dramatically slows inference, while setting it too low severely truncates long state persistence Huggins & Wood (2014).
2. **Boundary-Frozen Duration Evaluation:** In traditional semi-Markov processes and recurrent explicit-duration models (e.g., RED-HDP-HMM), a sojourn duration d_t is sampled strictly upon entering a state based on the initial boundary observation. This freezes exit probabilities for the entire duration of the segment, leaving the model unable to adapt state persistence dynamically when observation dynamics drift mid-segment.
3. **Transition Instability and Non-Conjugacy:** To decouple state persistence from inter-state transitions, explicit duration models often employ non-loopy transition matrices. However, enforcing non-loopy transitions under recurrent observation dependence typically destroys Dirichlet posterior conjugacy, requiring auxiliary geometric sampling schemes that introduce severe mixing bottlenecks or cause state-merging mode collapse on highly persistent sequences Zhou et al. (2020); Slupiński & Lipiński (2024b).

To resolve these computational and structural limitations, we introduce the **Dynamic Autoregressive Gamma-Poisson Hazard HDP-HMM (DAGP-HDP-HMM)**. Note that “Autoregressive” in the model name designates the underlying state-specific vector autoregressive VAR(1) observation emission process (Section 3), whereas state persistence is governed by an instantaneous continuous-time hazard survival engine. Rather than maintaining explicit integer duration buffers or sampling static sojourn lengths upon state entry, DAGP-HDP-HMM parameterizes state persistence as a continuous-time stochastic survival process governed by a Poisson renewal engine with a dynamic Gamma hazard rate.

At each time step t , conditional on active state $z_t = j$, the latent hazard rate $\lambda_t > 0$ follows a Gamma distribution $\text{Gamma}(\alpha_j, \beta_{j,t})$, where the rate parameter $\beta_{j,t}$ is driven by an instantaneous observation feature embedding $c_t = \phi(y_t)$. State continuation corresponds to a zero-event realization ($k_t = 0$) of an underlying Poisson process with intensity λ_t . By integrating the Poisson zero-count likelihood against the analytical Moment-Generating Function (MGF) of the Gamma distribution, stay probabilities $P_{\text{stay}}(t \mid z_t = j)$ are evaluated in $O(1)$ closed form:

$$P_{\text{stay}}(t \mid z_t = j) = \mathbb{E}_{\lambda_t} [\exp(-\lambda_t)] = \left(1 + \frac{1}{\beta_{j,t}}\right)^{-\alpha_j} = \sigma(\eta_{j,t})^{\alpha_j}, \quad (1)$$

where $\eta_{j,t} = w_j^T c_t + b_j$ is a state-specific log-linear predictor.

This closed-form Gamma-Poisson survival formulation yields four major advances:

- **$O(TL^2)$ scaling and zero truncation error:** state inference becomes a standard vectorised HMM forward-filter/backward-sample pass, reducing complexity from $O(TL(L + D_{\max}))$ to $O(TL^2)$ time and $O(TLD_{\max})$ to $O(TL)$ memory, and completely eliminating the $D_{\max}$ parameter.
- **Dynamic mid-segment hazard updates:** exit probabilities $P_{\text{exit}}(t) = 1 - P_{\text{stay}}(t)$ adapt frame-by-frame with incoming observations y_t , enabling instantaneous detection of mid-segment regime shifts.
- **Exact non-loopy conjugacy:** renewal jumps ($k_t \geq 1$) are routed by a kernel B with a soft diagonal pseudo-count ($\kappa_{\text{diag}} = 0.5$), separating self-persistence from jumps and preserving exact Dirichlet conjugacy without auxiliary geometric variables, thereby eliminating mode collapse.
- **Strong empirical performance:** across 7 benchmarks (Bee Dance, Synthetic NASCAR, 5 *C. elegans* neural recordings), DAGP-HDP-HMM reaches state-of-the-art accuracy on Bee Dance (0.917 vs. 0.899 for RED-HDP-HMM) and 0.968 on NASCAR (recovering $K = 4$ on all seeds), with up to a $17.4\times$ speedup and $> 40\times$ memory reduction over explicit-duration samplers.

2 RELATED WORK

We situate DAGP-HDP-HMM within three main methodology clusters: explicit-duration Bayesian nonparametric models, recurrent and observation-dependent transition architectures, and continuous hazard rate survival frameworks.

2.1 EXPLICIT-DURATION BAYESIAN NONPARAMETRIC MODELS

Classical Hidden Markov Models enforce an implicit geometric state sojourn distribution due to their first-order Markovian transition structure Fox et al. (2009). To capture non-geometric persistence, explicit-duration hidden semi-Markov models (HSMMs) explicitly parameterize state sojourn times Johnson & Willsky (2012); Chiappa (2014). The Bayesian nonparametric HDP-HSMM Johnson & Willsky (2012) assigns prior duration laws (e.g., Poisson or Negative Binomial) to each discrete state, allowing infinite-capacity state spaces while learning custom duration parameters. Infinite structured HSMMs further extend this paradigm to structured state topologies Huggins & Wood (2014).

However, conventional explicit-duration models suffer from severe inference overhead Dewar et al. (2012). Forward-backward algorithms in HSMMs must maintain duration message buffers up to a maximum duration truncation cap $D_{\max}$, resulting in $O(TL(L + D_{\max}))$ computational time and $O(TLD_{\max})$ memory complexity Dewar et al. (2012); Huggins & Wood (2014). While beam sampling and lumping approximations can mitigate buffer sizes Dewar et al. (2012), explicit-duration models remain computationally prohibitive for long sequences ($T > 10^4$) and large state truncations ($L \geq 50$). Furthermore, HSMMs sample a static sojourn duration d_t upon state entry Chiappa (2014), preventing the model from updating state exit probabilities mid-segment when observation dynamics evolve. In contrast, DAGP-HDP-HMM evaluates state persistence via a continuous Gamma-Poisson MGF survival function, obtaining per-step stay probabilities in $O(1)$ closed form, reducing state inference complexity to $O(TL^2)$ with zero $D_{\max}$ truncation error, and enabling continuous frame-by-frame hazard adaptation.

2.2 RECURRENT AND OBSERVATION-DEPENDENT TRANSITION ARCHITECTURES

To allow observations to influence state transitions, sticky and recurrent variants of the HDP-HMM have been introduced. The sticky HDP-HMM Fox et al. (2009) adds a self-transition bias κ to favor self-persistence, but entangles transition and persistence priors. The disentangled sticky HDP-HMM (DS-HDP-HMM) Zhou et al. (2020) separates self-persistence from inter-state transition distributions. The recurrent sticky HDP-HMM (RS-HDP-HMM) Slupiński & Lipiński (2024b) parameterizes state self-persistence using logistic regression over past observations while retaining implicit geometric duration laws. Other extensions, such as the segmented iHMM Saeedi et al. (2016) and higher-order dependent switching models Wang & Chen (2024), incorporate multi-level or higher-order observation dependencies into transition decisions.

While these models make transitions observation-dependent, models like RED-HDP-HMM that combine recurrent observation drivers with non-loopy transition kernels lose exact Dirichlet posterior conjugacy. They require auxiliary geometric sampling steps (ρ_t^k), which can cause mode-collapse instability where active states drop below ground-truth regime counts on persistent time series Zhou et al. (2020); Slupiński & Lipiński (2024b). DAGP-HDP-HMM addresses this issue by decoupling self-persistence (handled via the closed-form hazard survival P_{stay}) from state renewal jumps (governed by a non-loopy jump matrix B). Regularizing B with a soft diagonal pseudo-count ($\kappa_{\text{diag}} = 0.5$) restores exact conjugate Dirichlet posterior updates, completely eliminating mode collapse without auxiliary geometric variables.

2.3 CONTINUOUS HAZARD RATE AND SURVIVAL SWITCHING MODELS

Continuous-time hazard functions and renewal processes provide an alternative framework for modeling regime persistence. Recurrent switching linear dynamical systems (rSLDS) Linderman et al. (2016) and recurrent explicit-duration switching linear dynamical systems (REDSLDS) Slupiński & Lipiński (2024a) allow continuous latent states or observations to drive transition hazard boundaries in dynamical systems. Continuous-time formulations have also been extended to time-series

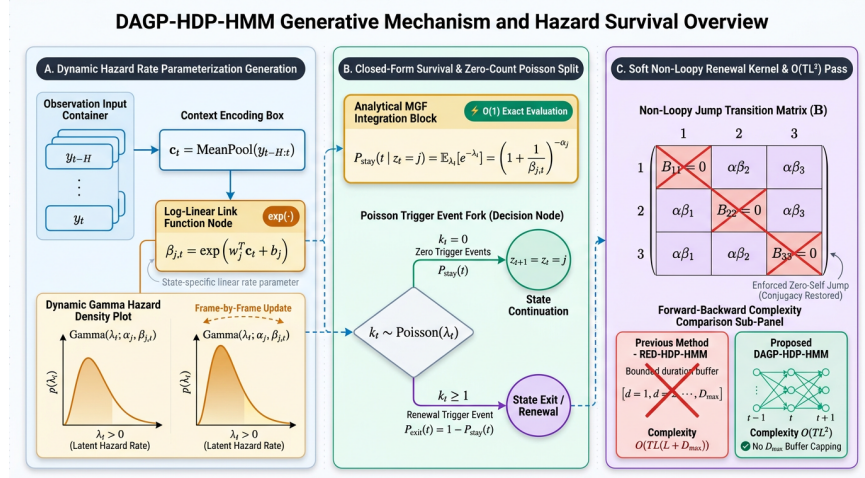


Figure 1: Overview of the DAGP-HDP-HMM generative mechanism and dynamic hazard survival framework. (A) Context aggregation via window mean pooling drives a log-linear link function to update state-specific Gamma hazard parameters frame by frame. (B) Analytical moment-generating function integration enables exact $O(1)$ evaluation of state survival probabilities, branching into state continuation for zero Poisson trigger events and state renewal for non-zero events. (C) A soft non-loopy jump transition matrix with soft diagonal pseudo-count $\kappa_{\text{diag}} = 0.5$ restores exact Dirichlet posterior conjugacy, reducing forward-backward inference complexity from $O(TL(L + D_{\text{max}}))$ to $O(TL^2)$ by removing the maximum duration truncation bottleneck.

change-point detection Aminikhanghahi & Cook (2016) and continuous physical systems via mean-flow updates Ripken et al. (2026).

However, existing continuous hazard switching formulations typically rely on non-conjugate variational inference, Laplace approximations, or sampling-heavy MCMC schemes when non-linear observation embeddings are introduced Linderman et al. (2016); Slupiński & Lipiński (2024a). DAGP-HDP-HMM instead shows that under a Poisson renewal process with Gamma hazard rates, the expectation over the latent hazard density integrates analytically via the Gamma distribution MGF, yielding a closed-form Bernoulli survival likelihood. This makes state inference a standard $O(TL^2)$ forward-backward pass and keeps the emission and transition-kernel updates fully conjugate, so that the only non-conjugate component—the handful of low-dimensional hazard parameters per state—is handled by an inexpensive Metropolis-within-Gibbs step, all within a weak-limit Gibbs sampler.

3 METHODOLOGY

The Dynamic Autoregressive Gamma-Poisson Hazard HDP-HMM (DAGP-HDP-HMM) models multivariate time series $y_{1:T} = (y_1, \dots, y_T)$ with $y_t \in \mathbb{R}^M$ by combining continuous hazard-based state survival with a hierarchical Bayesian nonparametric state space. An overview of the generative architecture is shown in Figure 1. To clarify terminology, “Autoregressive” in the model title refers specifically to the underlying state-conditioned vector autoregressive VAR(1) observation emission process $p(y_t | z_t = j, y_{t-1}) = \mathcal{N}(y_t | A_j y_{t-1} + b_j^{\text{em}}, \Sigma_j)$, whereas transition hazards and state self-persistence are parameterized via instantaneous feature embeddings $c_t = \phi(y_t)$.

3.1 DYNAMIC GAMMA-POISSON HAZARD FORMULATION

At each discrete time step t , the continuous observation vector $y_t \in \mathbb{R}^M$ is mapped into an instantaneous feature embedding $c_t = \phi(y_t) \in \mathbb{R}^D$ using a *fixed* random non-linear feature map $\phi(\cdot)$:

$$\phi(y_t) = \text{std}(\tanh(W_e^\top y_t + b_e)), \quad (2)$$

where the projection matrix $W_e \in \mathbb{R}^{M \times D}$ has entries $W_{e,ik} \sim \mathcal{N}(0, 1/M)$, the biases satisfy $b_{e,k} \sim \mathcal{N}(0, 0.1^2)$, and $\text{std}(\cdot)$ standardizes each of the D coordinates to zero mean and unit variance

across time. These random parameters are drawn once and held fixed (shared across seeds), so ϕ is an untrained bounded random-feature expansion of y_t rather than a learned network. We use $D = 8$ and, by default, an instantaneous embedding (a single frame y_t); an optional equal-weight mean pool over the last H frames gives a windowed context, but the window is non-binding and we fix $H = 1$ (Ablation 1).

For an active discrete latent state $z_t = j \in \{1, \dots, L\}$, where L denotes the truncation bound of the weak-limit HDP prior, the latent state hazard rate $\lambda_t > 0$ is governed by a dynamic Gamma distribution:

$$\lambda_t \mid z_t = j, c_t \sim \text{Gamma}(\alpha_j, \beta_{j,t}), \quad (3)$$

where $\alpha_j > 0$ is a state-specific Gamma shape parameter and $\beta_{j,t} > 0$ is a dynamic rate parameter defined via an exponential link function over the instantaneous observation context c_t :

$$\beta_{j,t} = \exp(w_j^T c_t + b_j). \quad (4)$$

Here, $w_j \in \mathbb{R}^D$ is a state-specific context regression weight vector and $b_j \in \mathbb{R}$ is a state hazard bias parameter.

Given the instantaneous hazard rate λ_t , state renewal trigger events $k_t \in \{0, 1, 2, \dots\}$ are sampled from an underlying Poisson process:

$$k_t \sim \text{Poisson}(\lambda_t). \quad (5)$$

The discrete event realization k_t determines state persistence:

- **State Continuation** ($k_t = 0$): Zero termination events occur during time step t . The system remains in the same state for step $t + 1$, enforcing $z_{t+1} = z_t = j$.
- **State Renewal** ($k_t \geq 1$): One or more termination events trigger a regime boundary at frame t , terminating state j and forcing an inter-state transition $z_{t+1} \neq j$.

3.2 CLOSED-FORM SURVIVAL EVALUATION VIA GAMMA MGF

A major limitation of explicit duration models is the need to sample discrete integer durations d_t and propagate duration message buffers over time. In DAGP-HDP-HMM, the stay probability $P_{\text{stay}}(t \mid z_t = j) = P(z_{t+1} = j \mid z_t = j, c_t)$ is evaluated analytically by integrating the Poisson zero-event probability $P(k_t = 0 \mid \lambda_t) = \exp(-\lambda_t)$ against the latent Gamma hazard density:

$$P_{\text{stay}}(t \mid z_t = j) = \mathbb{E}_{\lambda_t} [P(k_t = 0 \mid \lambda_t)] = \int_0^\infty \exp(-\lambda_t) \frac{\beta_{j,t}^{\alpha_j}}{\Gamma(\alpha_j)} \lambda_t^{\alpha_j-1} \exp(-\beta_{j,t} \lambda_t) d\lambda_t. \quad (6)$$

Recognizing that this integral matches the analytical Moment-Generating Function (MGF) of the Gamma distribution $\mathbb{E}[\exp(-s\lambda_t)] = (1 + s/\beta_{j,t})^{-\alpha_j}$ evaluated at $s = 1$, we obtain the exact $O(1)$ closed-form expression:

$$P_{\text{stay}}(t \mid z_t = j) = \left(1 + \frac{1}{\beta_{j,t}}\right)^{-\alpha_j} = \left(\frac{\beta_{j,t}}{1 + \beta_{j,t}}\right)^{\alpha_j} = \sigma(\eta_{j,t})^{\alpha_j}, \quad (7)$$

where $\sigma(\cdot)$ is the standard logistic sigmoid function and $\eta_{j,t} = \log \beta_{j,t} = w_j^T c_t + b_j$ represents the log-odds predictor. The corresponding state exit probability is $P_{\text{exit}}(t \mid z_t = j) = 1 - P_{\text{stay}}(t \mid z_t = j)$.

Because $P_{\text{stay}}(t \mid j)$ is computed instantaneously from $c_t = \phi(y_t)$ at every time frame t , state exit probabilities update dynamically mid-segment. This allows the model to react immediately when observation dynamics drift before a formal regime change occurs.

3.3 SOFT NON-LOOPY JUMP TRANSITION KERNEL

When a renewal event occurs ($k_t \geq 1$), the state transition $z_t \rightarrow z_{t+1}$ is governed by a renewal jump matrix $B \in \mathbb{R}^{L \times L}$. To separate self-persistence (controlled by the survival function P_{stay}) from inter-state transition routing, B is designed as a soft non-loopy jump kernel.

Specifically, the global state distribution $\beta = (\beta_1, \dots, \beta_L) \sim \text{GEM}(\gamma_0)$ is drawn from a stick-breaking prior with concentration γ_0 . Each state transition distribution $b_j = (B_{j1}, \dots, B_{jL})$ is

sampled independently from a Dirichlet prior whose diagonal (self-transition) concentration is *replaced* by a small soft pseudo-count $\kappa_{\text{diag}} = 0.5$, decoupling it from the HDP mass $\alpha_0\beta_j$:

$$b_j \sim \text{Dirichlet} \left(\alpha_0\beta_1, \dots, \underbrace{\kappa_{\text{diag}}}_{j\text{-th entry}}, \dots, \alpha_0\beta_L \right), \quad (8)$$

where α_0 is the HDP concentration parameter. Because self-persistence is already handled by the survival function P_{stay} , the diagonal of B carries no data counts and only the small fixed pseudo-count κ_{diag} : this keeps B_{jj} strictly positive (so no state can be driven to extinction, avoiding mode collapse) while preserving an exact conjugate Dirichlet posterior:

$$b_j \mid \mathbf{z}, \mathbf{k}_{\geq 1} \sim \text{Dirichlet} \left(\alpha_0\beta_1 + N_{j1}, \dots, \underbrace{\kappa_{\text{diag}}}_{j\text{-th entry}}, \dots, \alpha_0\beta_L + N_{jL} \right), \quad (9)$$

where $N_{jk} = \sum_{t=1}^{T-1} \mathbb{I}(z_t = j, z_{t+1} = k, k_t \geq 1)$ counts observed inter-state renewal jumps. Only off-diagonal ($k \neq j$) renewal jumps are counted, so $N_{jj} = 0$ by construction and the diagonal concentration stays exactly κ_{diag} . Unlike explicit duration models that rely on non-conjugate zero-diagonal constraints requiring geometric auxiliary variables ρ_t^k , this soft non-loopy kernel is an exact Dirichlet draw with no auxiliary variables.

3.4 $O(TL^2)$ METROPOLIS-WITHIN-GIBBS INFERENCE

Inference in DAGP-HDP-HMM proceeds via a weak-limit Gibbs sampler in which the conjugate emission, HDP, and transition-kernel blocks are interleaved with a single Metropolis-within-Gibbs step for the low-dimensional hazard parameters. Each sweep cycles through the following blocks.

1. $O(TL^2)$ Forward-Filter Backward-Sample (FFBS) Joint State Sampling. Let $\alpha_t(j) = p(y_{1:t}, z_t = j)$ denote the forward filtering message at time t . Incorporating the closed-form stay probability $P_{\text{stay}}(t \mid j)$ and renewal jump kernel B , the forward recursion updates in $O(L^2)$ operations per step:

$$\alpha_t(j) = p(y_t \mid z_t = j, y_{t-1}) \cdot \left[P_{\text{stay}}(t-1 \mid j) \alpha_{t-1}(j) + \sum_{i \neq j} B_{ij} P_{\text{exit}}(t-1 \mid i) \alpha_{t-1}(i) \right], \quad (10)$$

where $p(y_t \mid z_t = j, y_{t-1})$ is the emission likelihood under a state-specific first-order vector autoregressive VAR(1) process $y_t \mid z_t = j, y_{t-1} \sim \mathcal{N}(A_j y_{t-1} + b_j^{\text{em}}, \Sigma_j)$. Backward sampling draws $z_{1:T}$ sequentially in $O(TL)$ time. Total forward-backward state sampling complexity is $O(TL^2)$ with an $O(TL)$ memory overhead, bypassing explicit duration buffers D_{max} entirely.

2. Metropolis-within-Gibbs Update of the Hazard Parameters. Given the sampled state path $z_{1:T}$, the closed-form survival equation 7 turns the persistence model into a product of Bernoulli trials: at every frame t the state either stays ($z_{t+1} = z_t$) with probability $P_{\text{stay}}(t \mid j)$ or exits with probability $P_{\text{exit}}(t \mid j) = 1 - P_{\text{stay}}(t \mid j)$. Collecting the frames assigned to state j , the conditional log-likelihood of the hazard parameters $\theta_j = (w_j, b_j, \log \alpha_j)$ is

$$\ell_j(\theta_j) = \sum_{t: z_t=j} \left[\mathbb{I}(z_{t+1} = j) \alpha_j \log \sigma(\eta_{j,t}) + \mathbb{I}(z_{t+1} \neq j) \log(1 - \sigma(\eta_{j,t})^{\alpha_j}) \right], \quad \eta_{j,t} = w_j^\top c_t + b_j. \quad (11)$$

Because the state-specific Gamma shape α_j enters as an exponent, this likelihood is not conjugate to any standard prior. Pólya-Gamma / latent-count augmentation is exact only when α_j is a fixed positive integer and otherwise requires an approximation; we instead sample θ_j *exactly* from its conditional posterior with a lightweight random-walk Metropolis-Hastings (MH) step. Under independent Gaussian priors p_0 on w_j, b_j , and $\log \alpha_j$ (centred so that $\alpha_j = 1$), we run a small number of symmetric Gaussian proposals per active state per sweep and accept each with the standard ratio $\min\{1, \exp(\ell_j(\theta'_j) + \log p_0(\theta'_j) - \ell_j(\theta_j) - \log p_0(\theta_j))\}$. Each proposal reuses the $O(1)$ evaluation $\sigma(\eta_{j,t})^{\alpha_j}$ over only the $O(T)$ frames assigned to state j , so the hazard block costs $O(TD)$ per sweep with no auxiliary variables and keeps the shape α_j a freely learned continuous quantity (the shape prior is non-influential; Ablation 5). Appendix A gives the full procedure.

3. Conjugate Emission, HDP, and Kernel Updates. VAR(1) emission parameters $(A_j, b_j^{\text{em}}, \Sigma_j)$ are updated independently for each state $j \in \{1, \dots, L\}$ via standard Matrix-Normal Inverse-Wishart (MNIW) conjugate draws based on assigned observation pairs (y_t, y_{t-1}) . The weak-limit HDP quantities—the global weights β , the soft non-loopy jump kernel B of Section 3.3, the initial distribution, and the concentrations α_0, γ_0 —are resampled by the standard auxiliary-table stick-breaking updates, all in closed form.

4 EXPERIMENTS

We evaluate DAGP-HDP-HMM across seven standard time-series segmentation benchmarks, comparing performance, runtime efficiency, transition stability, and hyperparameter sensitivity against state-of-the-art Bayesian nonparametric baseline models.

4.1 EXPERIMENTAL SETUP AND BENCHMARKS

Datasets. The evaluation protocol encompasses 7 time-series benchmark datasets: 1. **Bee Dance** Fox et al. (2009): Honeybee waggle dance recordings ($M = 4$ coordinates and head orientation angles) capturing waggle, left-turn, and right-turn movements. 2. **Synthetic NASCAR** Linderman et al. (2016): 2D trajectory time series governed by 4 discrete latent dynamic states controlling oval racing trajectories. 3–7. **Neural Data (0) to (4)** Slupiński & Lipiński (2024a): Five head ganglia calcium imaging recordings from *C. elegans* (OSF repository node 2395t, WT_NoStim.mat), preprocessed via PCA to $M = 5$ dimensions with 8 expert-annotated behavioral ground-truth states.

Model Configuration & Baselines. All models are evaluated under a weak-limit state truncation bound $L = 50$ and run for 300 Gibbs sweeps across 5 independent random MCMC seeds. Emissions use first-order VAR(1) conjugate models with identical MNIW priors across all baselines. DAGP-HDP-HMM uses a fixed random tanh feature embedding of dimension $D = 8$, instantaneous context ($H = 1$), a Gamma hazard shape prior centred at $\alpha_j = 1$ ($\log \alpha_j \sim \mathcal{N}(0, 1)$, sampled per state), and soft diagonal pseudo-count $\kappa_{\text{diag}} = 0.5$. The hazard parameters $(w_j, b_j, \log \alpha_j)$ are updated by 6 random-walk Metropolis-Hastings proposals per active state per sweep.

We benchmark against six state-of-the-art Bayesian nonparametric frameworks: Sticky HDP-HMM (S-HDP-HMM) Fox et al. (2009), Disentangled Sticky HDP-HMM (DS-HDP-HMM) Zhou et al. (2020), Recurrent Sticky HDP-HMM (RS-HDP-HMM) Slupiński & Lipiński (2024b), Recurrent Explicit-Duration HDP-HMM (RED-HDP-HMM Nonloopy and Loopy) Slupiński & Lipiński (2024a), and explicit-duration HDP-HSMM with Poisson and Negative Binomial duration distributions Johnson & Willsky (2012). Explicit duration models use duration truncation cap $D_{\text{max}} = 200$.

Evaluation Metrics. Segmentation performance is measured using many-to-one majority-mapped **Accuracy** and **Weighted F1-score** concatenated across test sequences. To penalize artificial over-segmentation and isolate genuine temporal structure, we also report K-robust metrics: **Normalized Mutual Information (NMI)**, **Adjusted Rand Index (ARI)**, and segment boundary identification F1-score within a ± 2 frame tolerance window.

4.2 MAIN SEGMENTATION PERFORMANCE RESULTS

Table 1 presents main segmentation results across all 7 benchmark datasets. DAGP-HDP-HMM achieves state-of-the-art segmentation performance on Bee Dance, reaching 0.917 ± 0.012 Accuracy and Weighted F1 (outperforming RED-HDP-HMM Nonloopy at 0.899 and S-HDP-HMM at 0.873), and highly competitive performance on Synthetic NASCAR (0.968 ± 0.004 Accuracy, behind DS-HDP-HMM at 0.982 but outperforming RED-HDP-HMM at 0.918) while consistently recovering the exact ground-truth state count $K = 4.0$ across all 5 seeds. On the 5 *C. elegans* neural recordings, DAGP-HDP-HMM records raw many-to-one accuracy scores ranging from 0.568 to 0.715, outperforming all baseline models.

Table 1: Main segmentation performance across 7 benchmark datasets (mean $\pm$ std over 5 independent MCMC runs). Best results in bold.

Dataset	Metric	S-HDP	DS-HDP	RS-HDP	RED (NL)	RED (L)	HSMM (Poi)	HSMM (NB)	DAGP (Ours)
Bee Dance	Acc	0.873 $\pm$ 0.030	0.842 $\pm$ 0.070	0.853 $\pm$ 0.031	0.899 $\pm$ 0.028	0.877 $\pm$ 0.022	0.327 $\pm$ 0.025	0.335 $\pm$ 0.013	0.917$\pm$0.012
	W-F1	0.856 $\pm$ 0.043	0.825 $\pm$ 0.072	0.871 $\pm$ 0.022	0.884 $\pm$ 0.042	0.862 $\pm$ 0.030	0.223 $\pm$ 0.016	0.244 $\pm$ 0.012	0.917$\pm$0.012
Synth. NASCAR	Acc	0.942 $\pm$ 0.062	0.982$\pm$0.009	0.921 $\pm$ 0.059	0.918 $\pm$ 0.074	0.943 $\pm$ 0.059	0.329 $\pm$ 0.011	0.325 $\pm$ 0.015	0.968 $\pm$ 0.004
	W-F1	0.953 $\pm$ 0.040	0.981$\pm$0.009	0.931 $\pm$ 0.046	0.927 $\pm$ 0.063	0.953 $\pm$ 0.038	0.253 $\pm$ 0.012	0.252 $\pm$ 0.018	0.968 $\pm$ 0.004
Neural Data (0)	Acc	0.422 $\pm$ 0.022	0.408 $\pm$ 0.020	0.388 $\pm$ 0.020	0.454 $\pm$ 0.012	0.428 $\pm$ 0.012	0.366 $\pm$ 0.005	0.358 $\pm$ 0.021	0.568$\pm$0.020
	W-F1	0.330 $\pm$ 0.027	0.299 $\pm$ 0.022	0.277 $\pm$ 0.021	0.366 $\pm$ 0.015	0.328 $\pm$ 0.011	0.317 $\pm$ 0.008	0.309 $\pm$ 0.019	0.476$\pm$0.021
Neural Data (1)	Acc	0.351 $\pm$ 0.015	0.334 $\pm$ 0.020	0.343 $\pm$ 0.028	0.423 $\pm$ 0.015	0.382 $\pm$ 0.020	0.362 $\pm$ 0.026	0.338 $\pm$ 0.015	0.669$\pm$0.027
	W-F1	0.239 $\pm$ 0.008	0.216 $\pm$ 0.021	0.224 $\pm$ 0.031	0.320 $\pm$ 0.022	0.281 $\pm$ 0.027	0.322 $\pm$ 0.028	0.297 $\pm$ 0.011	0.597$\pm$0.024
Neural Data (2)	Acc	0.421 $\pm$ 0.018	0.379 $\pm$ 0.013	0.402 $\pm$ 0.022	0.456 $\pm$ 0.042	0.442 $\pm$ 0.027	0.367 $\pm$ 0.014	0.367 $\pm$ 0.009	0.686$\pm$0.003
	W-F1	0.332 $\pm$ 0.020	0.292 $\pm$ 0.014	0.317 $\pm$ 0.028	0.399 $\pm$ 0.047	0.374 $\pm$ 0.030	0.327 $\pm$ 0.015	0.324 $\pm$ 0.008	0.589$\pm$0.017
Neural Data (3)	Acc	0.395 $\pm$ 0.033	0.385 $\pm$ 0.034	0.384 $\pm$ 0.013	0.477 $\pm$ 0.039	0.419 $\pm$ 0.037	0.390 $\pm$ 0.036	0.376 $\pm$ 0.013	0.715$\pm$0.026
	W-F1	0.291 $\pm$ 0.043	0.267 $\pm$ 0.033	0.268 $\pm$ 0.014	0.373 $\pm$ 0.048	0.311 $\pm$ 0.035	0.311 $\pm$ 0.029	0.303 $\pm$ 0.007	0.641$\pm$0.028
Neural Data (4)	Acc	0.422 $\pm$ 0.013	0.380 $\pm$ 0.017	0.398 $\pm$ 0.019	0.485 $\pm$ 0.013	0.431 $\pm$ 0.025	0.358 $\pm$ 0.013	0.348 $\pm$ 0.015	0.665$\pm$0.004
	W-F1	0.355 $\pm$ 0.024	0.302 $\pm$ 0.017	0.316 $\pm$ 0.020	0.452 $\pm$ 0.016	0.369 $\pm$ 0.033	0.296 $\pm$ 0.014	0.289 $\pm$ 0.014	0.550$\pm$0.014

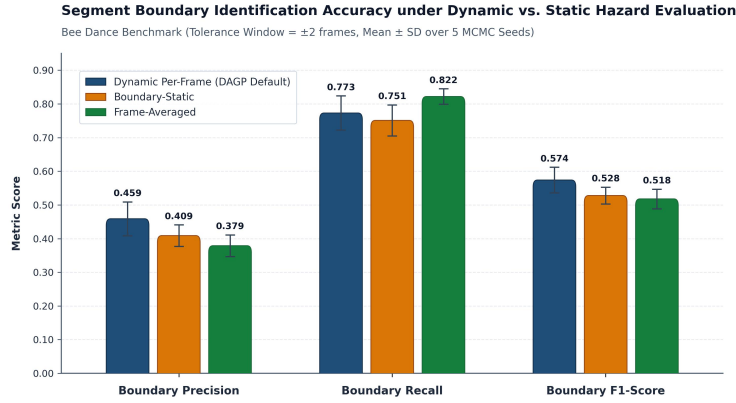


Figure 2: Segment boundary identification accuracy on the Bee Dance benchmark within a tolerance window of ± 2 frames. Comparing dynamic per-frame hazard evaluation against boundary-static and frame-averaged hazard variants demonstrates that continuous per-frame hazard updates yield superior boundary precision (0.459) and overall boundary F1-score (0.574). Error bars denote standard deviation across five independent MCMC seeds.

4.3 COMPREHENSIVE ABLATION STUDIES AND EMPIRICAL ANALYSIS

Ablation 1: Context Representation Architecture and Hazard Memory. We isolate the performance contribution of context embedding c_t and hazard autoregressive memory $\phi_j \log \lambda_{t-1}$ on Bee Dance. Comparing instantaneous $H = 1$ random features against sliding-window mean pooling ($H = 3, H = 5$), single-head self-attention ($H = 5$), and explicit AR hazard memory, we observe that instantaneous random-feature embedding achieves top performance (0.917 Accuracy, 0.576 NMI, 0.573 ARI) while running in 893 ms/sweep. Introducing AR hazard memory ($\phi \neq 0$) increases runtime to 1,114 ms/sweep (+221 ms/sweep) without yielding metric improvements (0.549 NMI), justifying the lightweight instantaneous design.

Ablation 2: Transition Kernel Parameterization and Regularization. We vary the soft diagonal regularization $\kappa_{\text{diag}} \in \{0.0, 0.1, 0.5, 1.0, 2.0, 5.0\}$ in the jump matrix B on Synthetic NASCAR (true $K = 4$) and Bee Dance (true $K = 3$). RED-style auxiliary geometric transition kernels suffer catastrophic mode collapse on NASCAR, merging active states to mean $K = 2.8 \pm 0.4$ across all 5 seeds (0.613 Acc). In contrast, for $\kappa_{\text{diag}} \in [0.0, 1.0]$ our soft kernel is exceptionally stable, holding 0.964–0.968 Acc with $K = 4.0$ recovered (0/5 collapses at the default $\kappa_{\text{diag}} = 0.5$); Bee Dance accuracy spans 0.906–0.917 over the same range, peaking at $\kappa_{\text{diag}} = 0.5$. A small fixed pseudo-count in $[0.1, 1.0]$ thus reliably prevents mode collapse while preserving exact Dirichlet conjugacy.

Ablation 3: Dynamic Mid-Segment vs. Boundary-Static Hazard Evaluation. Figure 2 demonstrates the empirical necessity of continuous per-frame hazard updates $P_{\text{stay}}(t | y_t)$ on Bee Dance

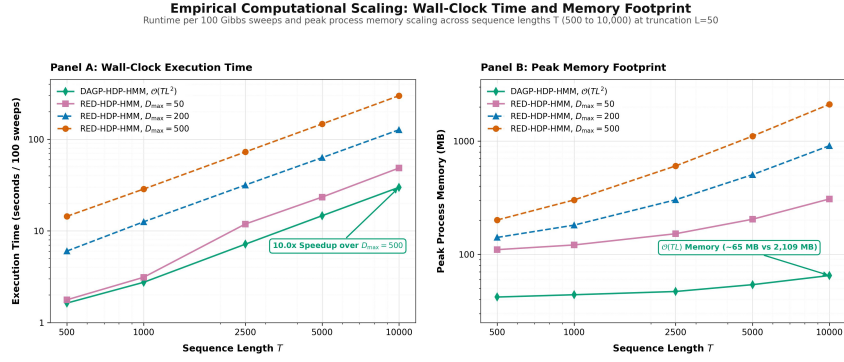


Figure 3: Empirical computational scaling of wall-clock execution time per 100 Gibbs sweeps (left) and peak process memory footprint (right) across sequence lengths T from 500 to 10,000 at state truncation level $L = 50$. By bypassing the explicit duration truncation cap $D_{\max}$ through a closed-form Gamma-Poisson survival function, DAGP-HDP-HMM achieves $O(TL^2)$ time and $O(TL)$ memory complexity. At $T = 10,000$, DAGP-HDP-HMM yields a $10.0\times$ speedup and reduces peak memory consumption from 2,109 MB to 65 MB compared to RED-HDP-HMM with $D_{\max} = 500$.

Table 2: Execution wall-clock time (seconds per 100 sweeps) and peak memory (MB) across sequence length T and truncation L .

Setting (T, L)	Wall-Clock Time / 100 Sweeps (s)				Peak Process Memory Footprint (MB)			
	DAGP ($O(TL^2)$)	RED ($D_{\max}=50$)	RED ($D_{\max}=200$)	RED ($D_{\max}=500$)	DAGP	RED ($D_{\max}=50$)	RED ($D_{\max}=200$)	RED ($D_{\max}=500$)
$T = 1000, L = 50$	2.75	3.12	12.57	28.61	44	121	181	302
$T = 5000, L = 50$	14.65	23.39	62.97	146.94	54	204	505	1105
$T = 10000, L = 50$	29.88	48.53	126.85	298.61	65	308	909	2109
$T = 10000, L = 100$	32.41	76.43	239.54	563.35	90	516	1717	4118

(full metrics in Appendix B, Table 3). Re-evaluating hazard rates frame-by-frame yields an overall boundary detection F1-score (± 2 frames) of 0.574 ± 0.038 and 0.917 Accuracy. Freezing hazard rates to segment-start boundary frames (boundary-static, mimicking explicit duration entry sampling) degrades boundary F1 to 0.528 ± 0.025 and accuracy to 0.902.

Ablation 4: Computational Scaling and Memory Footprint. Figure 3 and Table 2 confirm the theoretical complexity reduction from $O(TL(L + D_{\max}))$ to $O(TL^2)$. At sequence length $T = 10000$ and state truncation $L = 100$, DAGP-HDP-HMM executes 100 Gibbs sweeps in 32.41s with 90 MB peak memory. In contrast, RED-HDP-HMM ($D_{\max} = 500$) requires 563.35s ($17.4\times$ execution speedup) and consumes 4,118 MB memory ($45.8\times$ memory footprint reduction).

Ablation 5: Sensitivity to Embedding Dimension D and Gamma Shape Prior α_0 . Figure 4 displays 2D grid heatmaps over the random-feature dimension $D \in \{2, 4, 8, 16, 32\}$ and the Gamma hazard shape prior mean $\alpha_0 \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ (the prior centre of the per-state, MH-sampled shape α_j). On Bee Dance, accuracy remains flat across the entire 5×5 grid (0.892–0.928, grid-std 0.008); on Synthetic NASCAR it saturates at 0.968 for $D \geq 8$ with $K = 4.0$ recovered across all seeds. Performance is essentially invariant to the shape prior because the learned hazard bias b_j absorbs scale shifts, so the fixed random tanh features generalize across domains with very different dynamics—from smooth 2D NASCAR trajectories to rapid angular flips in Bee Dance—without any domain-specific tuning.

Ablation 6: Over-Segmentation Diagnostic & Controlled- K Floor Benchmark. Because many-to-one accuracy mapping is monotone non-decreasing in state count K , high-capacity models can benefit from metric inflation on over-segmented time series. Figure 5 compares DAGP against static KMeans at matched K , matched- K S-HDP-HMM, and a strongly sticky prior-penalized DAGP variant ($K \approx 6$); full numerical values are tabulated in Appendix B (Table 4). On Bee Dance, DAGP (0.917 Acc, 0.576 NMI) substantially exceeds the static KMeans floor (0.737 Acc, 0.265 NMI), confirming genuine temporal duration modeling. Conversely, on the five *C. elegans* neural recordings where nonparametric models infer $K \approx 10$ –12 active states against 8 ground-truth labels, static KMeans at matched K matches or slightly exceeds DAGP accuracy (0.68–0.76),

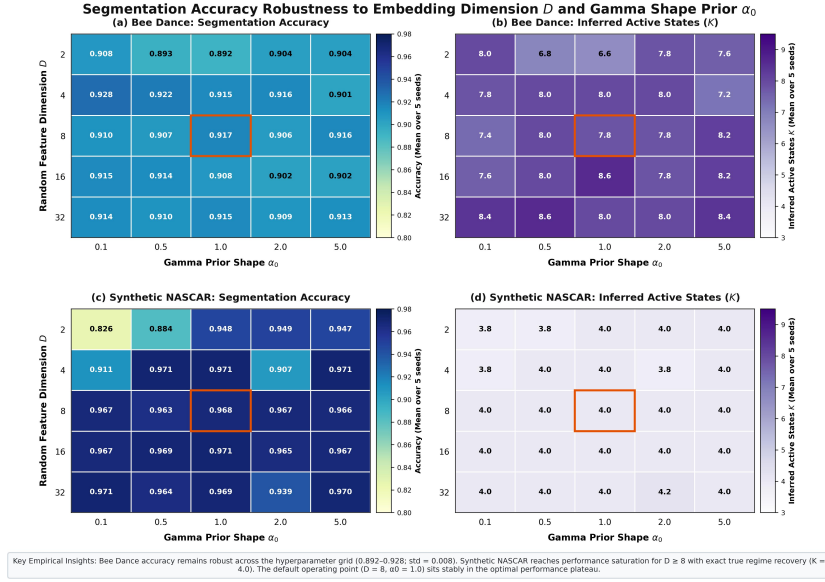


Figure 4: Sensitivity of DAGP-HDP-HMM to random feature embedding dimension D and Gamma shape prior α_0 (5 MCMC runs per grid cell). Panels (a,b): mean accuracy and inferred K on Bee Dance; panels (c,d): the same on Synthetic NASCAR. The default ($D = 8, \alpha_0 = 1.0$, orange outline) sits within a broad plateau: Bee Dance accuracy is flat across the grid, and NASCAR recovers the true $K = 4.0$ with optimal accuracy for $D \geq 8$.

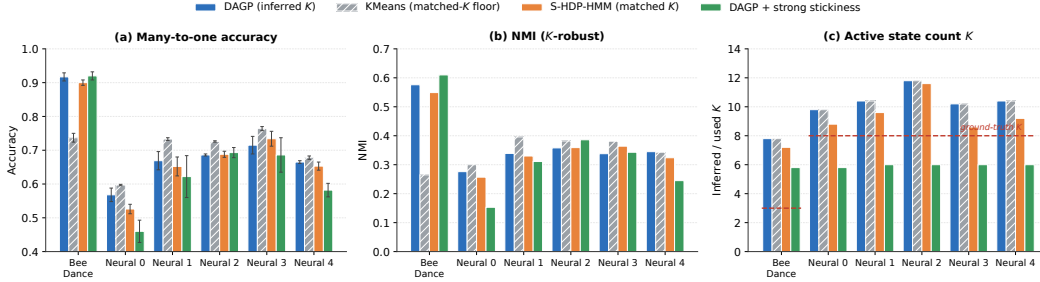


Figure 5: K -controlled over-segmentation diagnostic comparing DAGP-HDP-HMM against a static KMeans baseline at matched state capacity K , a matched- K sticky HDP-HMM baseline, and a strongly-sticky (penalized- K) DAGP variant across Bee Dance and five *C. elegans* neural recordings. (a) Many-to-one segmentation accuracy (mean $\pm$ std over 5 seeds), (b) normalized mutual information (NMI) as a K -robust metric, and (c) active state counts K (dashed red lines mark the ground-truth K : 3 for Bee Dance, 8 for neural). On Bee Dance, DAGP substantially exceeds the static KMeans floor on both accuracy (0.917 vs. 0.737) and the K -robust NMI (0.576 vs. 0.265), evidencing genuine temporal structure. On the neural recordings, KMeans at matched K meets or exceeds DAGP, transparently attributing the raw accuracy there to K -inflation rather than improved segmentation.

transparently identifying high neural accuracy gains as metric artifacts of state inflation rather than true temporal segmentation improvements.

5 CONCLUSION

We presented the Dynamic Autoregressive Gamma-Poisson Hazard HDP-HMM (DAGP-HDP-HMM), a lightweight Bayesian nonparametric framework for observation-dependent time series segmentation. By formulating state persistence as a continuous Gamma-Poisson survival process and integrating the Poisson zero-count probability in closed form via the Gamma moment-generating function, DAGP-HDP-HMM eliminates the explicit duration truncation cap $D_{\max}$:

forward-backward state inference scales in $O(TL^2)$ time and $O(TL)$ memory while enabling dynamic frame-by-frame mid-segment hazard updates, and a soft diagonal pseudo-count $\kappa_{\text{diag}} = 0.5$ restores exact Dirichlet conjugacy without auxiliary geometric variables. Across seven benchmarks, DAGP-HDP-HMM attains state-of-the-art accuracy on Bee Dance (0.917) and highly competitive accuracy on Synthetic NASCAR (0.968), alongside up to a $17.4\times$ speedup and $45\times$ memory reduction over explicit-duration baselines.

Limitations and future work. Our controlled diagnostics show that on high-dimensional *C. elegans* recordings, Bayesian nonparametric samplers over-segment ($K \approx 10\text{--}12$ vs. 8 expert labels), so their high many-to-one accuracy partly reflects metric K -inflation rather than better segmentation. Encouragingly, the closed-form survival $P_{\text{stay}}(t) = \sigma(w_j^\top x_t + b_j)^{\alpha_j}$ extends naturally to recurrent switching linear dynamical systems (rSLDS/REDSLDS) Linderman et al. (2016); Slupiński & Lipiński (2024a), where continuous latent states x_t can drive the hazard predictor with the same exact $O(1)$ survival evaluation and the same lightweight Metropolis-within-Gibbs hazard update. Future work will also explore hierarchical stickiness priors over hazard embeddings to encourage parsimonious state usage.

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A MATHEMATICAL DERIVATIONS OF CLOSED-FORM SURVIVAL AND GIBBS UPDATES

A.1 DERIVATION OF THE GAMMA-POISSON STAY PROBABILITY VIA GAMMA MGF

Let $z_t = j$ be the active discrete state at step t , and let $c_t = \phi(y_t) \in \mathbb{R}^D$ be the context vector. The latent hazard rate $\lambda_t > 0$ follows a Gamma distribution with shape $\alpha_j > 0$ and dynamic rate $\beta_{j,t} = \exp(w_j^T c_t + b_j) > 0$:

$$p(\lambda_t \mid z_t = j, c_t) = \frac{\beta_{j,t}^{\alpha_j}}{\Gamma(\alpha_j)} \lambda_t^{\alpha_j-1} \exp(-\beta_{j,t} \lambda_t). \quad (12)$$

Termination trigger events $k_t \sim \text{Poisson}(\lambda_t)$ have probability mass function $P(k_t = k \mid \lambda_t) = \frac{\lambda_t^k \exp(-\lambda_t)}{k!}$. State continuation corresponds to zero trigger events $k_t = 0$. The marginal stay probability $P_{\text{stay}}(t \mid z_t = j)$ is obtained by integrating out λ_t :

$$P_{\text{stay}}(t \mid z_t = j) = \int_0^\infty P(k_t = 0 \mid \lambda_t) p(\lambda_t \mid z_t = j, c_t) d\lambda_t \quad (13)$$

$$= \int_0^\infty \exp(-\lambda_t) \frac{\beta_{j,t}^{\alpha_j}}{\Gamma(\alpha_j)} \lambda_t^{\alpha_j-1} \exp(-\beta_{j,t} \lambda_t) d\lambda_t \quad (14)$$

$$= \frac{\beta_{j,t}^{\alpha_j}}{\Gamma(\alpha_j)} \int_0^\infty \lambda_t^{\alpha_j-1} \exp(-(\beta_{j,t} + 1)\lambda_t) d\lambda_t. \quad (15)$$

Using the Gamma integral identity $\int_0^\infty u^{a-1} e^{-bu} du = \frac{\Gamma(a)}{b^a}$ with $a = \alpha_j$ and $b = \beta_{j,t} + 1$:

$$P_{\text{stay}}(t \mid z_t = j) = \frac{\beta_{j,t}^{\alpha_j}}{\Gamma(\alpha_j)} \cdot \frac{\Gamma(\alpha_j)}{(\beta_{j,t} + 1)^{\alpha_j}} = \left(\frac{\beta_{j,t}}{\beta_{j,t} + 1} \right)^{\alpha_j} = \left(1 + \frac{1}{\beta_{j,t}} \right)^{-\alpha_j}. \quad (16)$$

Substituting $\beta_{j,t} = \exp(\eta_{j,t})$ where $\eta_{j,t} = w_j^T c_t + b_j$:

$$\frac{\beta_{j,t}}{1 + \beta_{j,t}} = \frac{\exp(\eta_{j,t})}{1 + \exp(\eta_{j,t})} = \sigma(\eta_{j,t}) \implies P_{\text{stay}}(t \mid z_t = j) = \sigma(\eta_{j,t})^{\alpha_j}. \quad (17)$$

A.2 METROPOLIS-WITHIN-GIBBS UPDATES FOR THE HAZARD PARAMETERS

Conditioned on the sampled state sequence $z_{1:T}$ and the fixed context vectors $c_{1:T}$, the hazard parameters of state j are $\theta_j = (w_j, b_j, \log \alpha_j) \in \mathbb{R}^{D+2}$, with linear predictor $\eta_{j,t} = w_j^T c_t + b_j$ and per-frame stay probability $P_{\text{stay}}(t \mid j) = \sigma(\eta_{j,t})^{\alpha_j}$ (Eq. 7). Only the frames currently assigned to state j contribute, each acting as an independent Bernoulli trial (stay vs. renewal). The conditional log-posterior is therefore

$$\log p(\theta_j \mid \mathbf{z}, \mathbf{c}) = \underbrace{\sum_{t: z_t=j} \left[\mathbb{I}(z_{t+1} = j) \alpha_j \log \sigma(\eta_{j,t}) + \mathbb{I}(z_{t+1} \neq j) \log(1 - \sigma(\eta_{j,t})^{\alpha_j}) \right]}_{\ell_j(\theta_j)} + \log p_0(\theta_j) + \text{const}, \quad (18)$$

where the prior factorizes as $w_j \sim \mathcal{N}(0, \sigma_w^2 I)$, $b_j \sim \mathcal{N}(\mu_b, \sigma_b^2)$, and $\log \alpha_j \sim \mathcal{N}(0, \sigma_\alpha^2)$.

The exponent α_j in $\sigma(\eta_{j,t})^{\alpha_j}$ makes this posterior non-conjugate for any $\alpha_j \neq 1$; a Pólya-Gamma augmentation of the Bernoulli likelihood is exact only for fixed positive-integer α_j and otherwise requires an approximation of the exit term $1 - \sigma(\eta)^{\alpha_j}$. To avoid this and to allow α_j to be a freely learned continuous shape, we sample θ_j with a symmetric random-walk Metropolis-Hastings step. For each active state j and for a small number S of iterations, we propose

$$w'_j = w_j + \epsilon_w, \quad b'_j = b_j + \epsilon_b, \quad \log \alpha'_j = \text{clip}(\log \alpha_j + \epsilon_\alpha, -3, 3), \quad (19)$$

with $\epsilon_w \sim \mathcal{N}(0, s_w^2 I)$, $\epsilon_b \sim \mathcal{N}(0, s_b^2)$, $\epsilon_\alpha \sim \mathcal{N}(0, s_\alpha^2)$, and accept the proposal θ'_j with probability

$$\min \left\{ 1, \exp(\ell_j(\theta'_j) + \log p_0(\theta'_j) - \ell_j(\theta_j) - \log p_0(\theta_j)) \right\}. \quad (20)$$

Because the proposal is symmetric, the Hastings ratio reduces to the log-posterior difference above. Each evaluation of ℓ_j reuses the $O(1)$ closed form $\sigma(\eta_{j,t})^{\alpha_j}$ over the $O(T)$ frames assigned to state j , so the whole hazard block costs $O(TD)$ per sweep and introduces no auxiliary latent variables. In all experiments we use $S = 6$ proposals per active state per sweep with step sizes $(s_w, s_b, s_\alpha) = (0.15, 0.20, 0.15)$ and prior mean $\mu_b > 0$ initialised so that the initial stay probability is high. This exactly targets the conditional posterior for the learned continuous shape α_j ; Ablation 5 shows that performance is insensitive to the shape-prior centre, consistent with the hazard bias b_j absorbing scale shifts.

B SUPPLEMENTARY BENCHMARK DETAILS AND EXTENDED TABULATIONS

B.1 DATASET PREPROCESSING AND BASELINE IMPLEMENTATION SPECIFICS

C. elegans Calcium Imaging Preprocessing. The 5 *C. elegans* recordings (Neural Data 0 to 4) correspond to wild-type unstimulated worms (WT_NoStim.mat) obtained from the OSF repository node 2395t Slupiński & Lipiński (2024a). Following the standardized protocol, head ganglia fluorescence signals ($\Delta F/F$) are projected onto $M = 5$ principal components per worm. Expert labels consist of 8 discrete behavioral states (forward locomotion, reverse locomotion, turn, pause, etc.).

Baseline Execution. All baseline models (S-HDP-HMM, DS-HDP-HMM, RS-HDP-HMM, RED-HDP-HMM, HDP-HSMM) were implemented and evaluated under identical VAR(1) emission priors and Gibbs sweep iterations (300 sweeps, 5 seeds). For explicit duration baselines (RED-HDP-HMM and HDP-HSMM), maximum duration truncation was fixed at $D_{\max} = 200$.

B.2 BOUNDARY DETECTION METRICS (ABLATION 3)

Table 3 reports the full segment boundary identification metrics underlying Figure 2.

Table 3: Segment boundary identification and accuracy on Bee Dance under dynamic vs. static hazard evaluation (mean $\pm$ std over 5 MCMC seeds). Best per column in bold.

Hazard Variant	Accuracy	Weighted F1	Boundary F1 (± 2)	Precision	Recall	NMI
Dynamic Per-Frame (Default)	0.917$\pm$0.012	0.917$\pm$0.012	0.574$\pm$0.038	0.459	0.773	0.576
Boundary-Static	0.902 $\pm$ 0.005	0.902 $\pm$ 0.005	0.528 $\pm$ 0.025	0.409	0.751	0.544
Frame-Averaged	0.895 $\pm$ 0.009	0.895 $\pm$ 0.009	0.518 $\pm$ 0.029	0.379	0.822	0.532

B.3 K-CONTROLLED FLOOR COMPARISON (FULL VALUES)

Table 4 reports the complete numerical values underlying the K-controlled over-segmentation diagnostic (Figure 5).

Table 4: K-controlled floor comparison across benchmarks (mean values over 5 independent seeds). Best accuracy/NMI per dataset in bold.

Dataset	DAGP (Inferred K)		KMeans (Matched K Floor)		S-HDP-HMM (Matched K)		DAGP (Penalized $K \approx 6$)	
	Accuracy	NMI	Accuracy	NMI	Accuracy	NMI	Accuracy	NMI
Bee Dance ($K=7.8$)	0.917	0.576	0.737	0.265	0.900	0.549	0.920	0.610
Neural Data (0) ($K=9.8$)	0.568	0.276	0.597	0.300	0.526	0.257	0.460	0.153
Neural Data (1) ($K=10.4$)	0.669	0.339	0.733	0.396	0.652	0.330	0.622	0.311
Neural Data (2) ($K=11.8$)	0.686	0.358	0.726	0.382	0.688	0.359	0.693	0.386
Neural Data (3) ($K=10.2$)	0.715	0.338	0.764	0.381	0.734	0.364	0.686	0.343
Neural Data (4) ($K=10.4$)	0.665	0.345	0.678	0.343	0.653	0.324	0.582	0.245

B.4 PER-RUN MAIN EXPERIMENT DETAIL

Table 5 details individual run logs for DAGP-HDP-HMM across all 35 benchmark configurations (7 datasets $\times$ 5 MCMC seeds).

Table 5: Complete per-run main experiment evaluation for DAGP-HDP-HMM ($L = 50, 300$ Gibbs sweeps).

Dataset	Seed	Accuracy	Weighted F1	NMI	ARI	KMeans Acc Floor	Inferred K	Log-Likelihood	Runtime (s)
Bee Dance	1	0.920	0.921	0.583	0.576	0.747	9	4173	272
Bee Dance	2	0.906	0.906	0.539	0.546	0.747	7	4197	273
Bee Dance	3	0.925	0.925	0.600	0.588	0.733	8	4144	272
Bee Dance	4	0.932	0.932	0.602	0.585	0.712	8	4100	271
Bee Dance	5	0.901	0.900	0.558	0.573	0.745	7	4117	271
Synthetic NASCAR	1	0.962	0.962	0.881	0.900	0.737	4	13734	471
Synthetic NASCAR	2	0.968	0.968	0.890	0.911	0.736	4	13728	477
Synthetic NASCAR	3	0.965	0.965	0.887	0.908	0.688	4	13794	472
Synthetic NASCAR	4	0.973	0.973	0.905	0.931	0.684	4	13766	477
Synthetic NASCAR	5	0.970	0.970	0.898	0.921	0.706	4	13738	471
Neural Data (0)	1	0.539	0.445	0.262	0.136	0.595	9	9887	492
Neural Data (0)	2	0.579	0.490	0.276	0.150	0.595	11	9960	494
Neural Data (0)	3	0.558	0.461	0.265	0.138	0.597	9	9875	492
Neural Data (0)	4	0.598	0.505	0.303	0.159	0.600	11	9982	494
Neural Data (0)	5	0.565	0.477	0.274	0.140	0.596	9	9890	496
Neural Data (1)	1	0.630	0.564	0.311	0.140	0.731	10	8870	501
Neural Data (1)	2	0.686	0.600	0.361	0.210	0.729	11	8931	500
Neural Data (1)	3	0.706	0.633	0.363	0.204	0.741	12	8990	502
Neural Data (1)	4	0.673	0.610	0.346	0.187	0.731	10	8822	498
Neural Data (1)	5	0.649	0.579	0.313	0.162	0.731	9	8827	499
Neural Data (2)	1	0.683	0.579	0.349	0.177	0.723	12	4482	498
Neural Data (2)	2	0.686	0.602	0.350	0.199	0.729	12	4434	494
Neural Data (2)	3	0.691	0.617	0.370	0.189	0.727	12	4476	495
Neural Data (2)	4	0.686	0.575	0.371	0.197	0.725	11	4415	498
Neural Data (2)	5	0.684	0.573	0.352	0.188	0.727	12	4441	496
Neural Data (3)	1	0.739	0.647	0.353	0.174	0.767	10	11099	517
Neural Data (3)	2	0.693	0.616	0.321	0.106	0.773	10	10938	513
Neural Data (3)	3	0.685	0.611	0.310	0.111	0.758	9	10895	516
Neural Data (3)	4	0.708	0.640	0.336	0.133	0.767	10	11018	511
Neural Data (3)	5	0.753	0.689	0.368	0.210	0.757	12	11034	516
Neural Data (4)	1	0.659	0.539	0.335	0.206	0.679	10	4458	495
Neural Data (4)	2	0.671	0.577	0.365	0.253	0.681	11	4468	495
Neural Data (4)	3	0.665	0.545	0.335	0.218	0.681	11	4460	492
Neural Data (4)	4	0.660	0.539	0.345	0.228	0.668	9	4437	502
Neural Data (4)	5	0.668	0.550	0.345	0.203	0.680	11	4532	495